Final Project: Financial Data Analysis

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Due on May 16, 2026 at 11:59 PM. No late submission is allowed.

1 Project

The attached data file contains the monthly returns of the 6 portfolios constructed by sorting stocks into different size and book-to-market ratio groups from Kenneth French's Data Library, covering the period from July 1926 to January 2026. In this project, you are required to explore the modeling of the monthly returns and explore trading strategies with the 6 portfolios.

2 Assessment Scheme

The report should include the following components:

- Introduction (15%)
  - Justify the motivation and introduce the research background.
  - Perform literature review related to the modeling of the monthly returns and relevant trading strategies.
  - Summarize the main contributions and highlight the key findings.
- Data Source and Processing (15%)
  - Describe the data source and the preprocessing procedures if any.
  - Summarize the data and describe the variables in the dataset.
- Modeling the Individual Portfolio Returns (30%)
  - Explore the distributional properties of the monthly returns of the 6 portfolios individually.
  - Explore the time series modeling (AR, MA, ARMA, ARIMA, GARCH, etc.) of the monthly returns of the 6 portfolios individually. Assess and interpret the model fit and the diagnostic plots.
  - Incorporate exogenous variables for the predictive modeling of the monthly returns of the 6 portfolios individually.
- Trading Strategies (20%)
  - Conduct joint multivariate time series modeling of the monthly returns of the 6 portfolios, test the cointegration relationships, and explore the profitability of statistical arbitrage strategies.
  - Conduct mean-variance analysis, plot the efficient frontier, backtest the optimal mean-variance strategy, and compare the results with the equally weighted portfolio.

  - Propose improvements to the plug-in mean-variance strategies and compare the results with the original strategies.

- Conclusions and References (10%)
  - Summarize the main findings and highlight the key contributions.
  - Attach the references in good format and cite them in the main body.

- The report should be well formatted and organized. (10%)

3 Submission

Submit a zip file containing the pdf file of report (the main body excluding the references and appendices should be no longer than 25 pages) and executable codes to Blackboard.